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Temperature gradient driven constituent redistribution in U–Zr alloys [1]

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Abstract

Although the phenomenon of constituent redistribution is common in U–Pu–Zr alloys irradiated under a wide range of conditions, it has been observed in U–Zr alloys only at elevated temperatures. Redistribution is relatively rapid and is essentially complete by 5 at% burnup. Experimental observations of constituent redistribution in U–Zr fuel elements are presented and analyzed. A model based on a thermal diffusion mechanism is proposed, and its computer implementation is described. The model calculations, supported by experimental observation, indicate that the excess enthalpy of solution of the bcc γ-phase controls the redistribution process as an additional driving force. A heat of transport of −50 to −100 kJ/mol in this phase results in the best match between calculation and experimental observations. The model predicts that constituent redistribution will be observed only when a region of the fuel operates at temperatures above 935 K.

1. Introduction

The thermal conductivity of uranium alloys is among the highest of nuclear reactor fuels, yet radial temperature gradients generated in rod-type fast reactor fuel elements appear to be sufficiently large to cause redistribution of the alloy constituents. This manifestation of thermotransport in nuclear fuel has been well characterized in other types of fuel rods, such as O and Pu thermo-transport in mixed oxide fuels [1]. The first observations of this phenomenon in metallic U–Pu–Zr fuel was reported by Murphy et al. [2] who identified three different concentric microstructural zones resulting from, primarily, radial Zr and U redistribution during irradiation. Renewed interest in metallic fast reactor fuel has yielded additional postirradiation data [3] that confirm the initial observation and show that substantial radial redistribution of U and Zr occurs at relatively short radiation exposures or burnup (Fig. 1). It is observed that the net migration of Zr in the ternary U–Pu–Zr alloy can be either up or down the temperature gradient depending on the local fuel temperature.

Constituent redistribution in metallic alloy fuels leads to radial changes in alloy phase composition and local power production. This, in turn, leads to variations in fission product generation, and irradiation behavior such as swelling, and mechanical and physical properties. Thus, understanding and predicting the constituent redistribution in metallic alloy fuels is of importance to the overall interpretation of fuel element behavior.

Others have attempted to model the constituent redistribution in metallic fuel. Ishida et al. [4] modeled redistribution in U–Pu–Zr; however, far-reaching assumptions and approximations are required in order to obtain a model usable in a fuel performance code. The reason for this is the paucity of thermodynamic and diffusion data for the ternary U–Pu–Zr alloy. To circumvent this problem, these researchers modeled the simpler and better characterized binary U–Zr alloy and attempted to extrapolate this work to the U–Pu–Zr system. Similarly, Ogawa et al. [5] developed a model for constituent redistribution in U–Zr alloys without attempting to extrapolate to the U–Pu–Zr system. The problem with these previous modeling attempts is that,

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until recently, constituent redistribution in U–Zr alloys had not been definitively observed. It was, therefore, impossible to validate the binary alloy redistribution model, making an extrapolation to the ternary alloy tenuous. This paper analyzes recent postirradiation data on U–Zr fuel, irradiated to 10 at% burnup at high temperature, in which constituent redistribution is definitely observed and characterized. A mechanistic thermo-transport model is proposed to describe the U–Zr redistribution phenomenon, which is incorporated into a computer subroutine capable of eventually interfacing with a general fuel performance code. Comparison of the model calculations with experimental data shows good agreement.

2. Experimental

The fuel elements in this experiment were part of the fuel development effort for the Integral Fast Reactor (IFR) program [3], and consisted of injection cast U–22(±1) at% Zr pins with a diameter of 4.34 mm and a length of 340 mm. These pins were clad in 600 mm long austenitic (D9) or ferritic (HT9) stainless steel tubes having diameters of 5.84 mm and wall thicknesses of 0.38 mm. The gap between the fuel pins and cladding is filled with sodium to provide a thermal bond. The irradiation experiment, designated subassembly X447, was performed in the Experimental Breeder Reactor II (EBR-II) at a maximum linear power of 33 000 W/m (330 W/cm) to a peak heavy metal burnup of 10 at%. The upward sodium coolant flow of 2.52 kg/s within the subassembly, with a core inlet temperature of 644 K, resulted in peak cladding temperatures in the range of 903 to 933 K occurring at the top of the fuel pin. Post-irradiation examinations were performed at 5 at% peak burnup, reached after ~ 284 effective full power days (EFPDs) and 10 at% after ~ 619 EFPDs. Neutron radiographs indicated radial zone formation near the top of the fuel pins (i.e., in essentially the highest temperature part of the fuel due to the high thermal conductivity of metallic fuel and the relatively flat axial power profile in EBR-II). These locations were selected for detailed metallography and electron microprobe analysis; two such sections are shown in Figs. 2 and 3. The similarities between the U–Zr samples and the U–Pu–Zr sample, shown in Fig. 1, are obvious. Both fuels developed a porous, Zr-enriched center and an apparently dense Zr-depleted intermediate zone. However, there are some key differences. The intermediate zone in U–Pu–Zr is completely depleted of the original high Zr bcc phase and indeed of nearly all Zr. A comparison of the amount of Zr lost from this zone and the

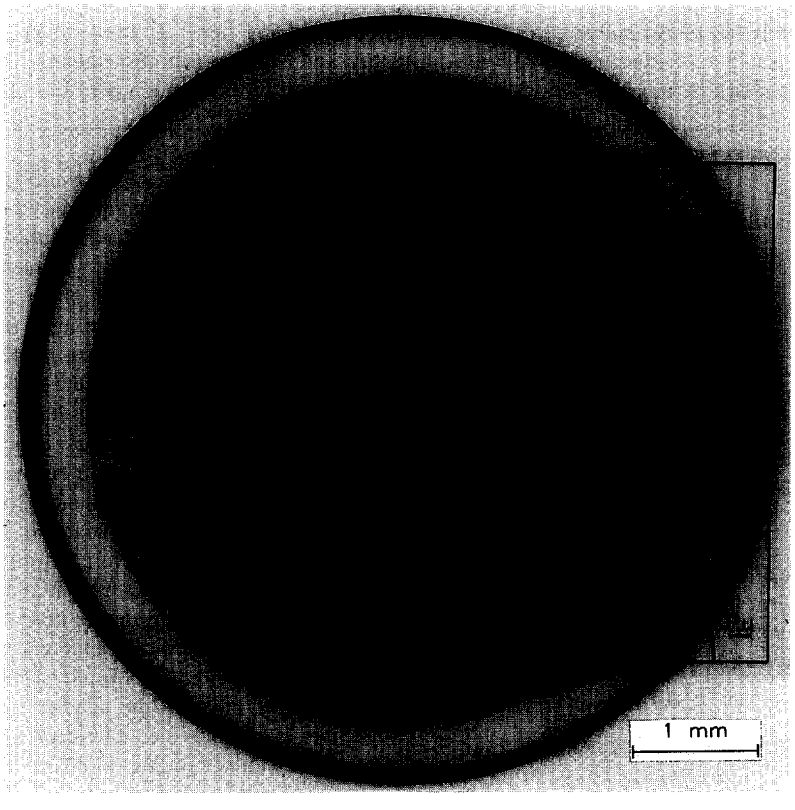


Fig. 1. Metallographic cross section with superimposed radial microprobe scans of element T179, U–19Pu–10Zr, at 0.22 m from bottom of fuel.

amount gained by the center zone leads to the conclusion that a significant fraction of Zr must also have migrated radially outward. Sections taken at elevations corresponding to lower fuel temperatures show depleted centers in U–Pu–Zr elements [3], which can only be explained by outward Zr migration (i.e., Zr migration down the temperature gradient). In contrast, evidence of outward Zr migration has not been found in the binary U–Zr fuel. In the latter, the depleted intermediate zone retains a fair amount of the high Zr-phase [which is the cause of the irregular Zr composition profiles observed in the binary fuel sections (Fig. 4) compared to those for the ternary fuel] and the amount of Zr lost from the depleted intermediate zone balances the amount gained in the enriched center.

Radial composition profiles were obtained with a defocused electron beam having a sample spot of 40 μm × 40 μm. In this way, average compositions were measured in the two-phase structure. Two examples are shown in Fig. 4; one taken at an elevation 32.6 cm from the fuel bottom ($z/L \approx 0.95$) of element designated DP-81 having a peak element burnup of 5 at%, and another taken at an elevation of 31.9 cm from the fuel bottom ($z/L \approx 0.93$) of the element designated DP-11 having a peak element burnup of 10 at%. The large irregularities in the scans are primarily due to the two-phase microstructure of the alloy. Absolute compositions in each of the three radial zones, given in Table 1, were quantified using U and Zr standards.

With the aid of the U–Zr phase diagram (Fig. 5), the composition profiles may be interpreted in the following way. By 5 at% burnup, the maximum Zr concentration in the center zone has increased from the initial 23 at% to approximately 45 at% and consists of a single-phase (γ-UZr). The Zr concentration in the intermediate two-phase zone has dropped commensurately resulting in a reduction of the high Zr γ-phase fraction. By 10 at% burnup, additional diffusion has diminished the Zr concentration gradient in the central zone, but the Zr concentration in the center zone has increased only slightly to 48 at%. As indicated in the phase diagram, 45 at% Zr roughly corresponds to the point where the γ-phase at the fuel center is in equilibrium with the γ_2-phase in the intermediate zone. Any further increase in Zr concentration at this location can only occur at a lower temperature. This postulated interpretation suggests the possibility of utilizing these zone boundaries as temperature markers, which would permit a more accurate reconstruction of the radial temper-

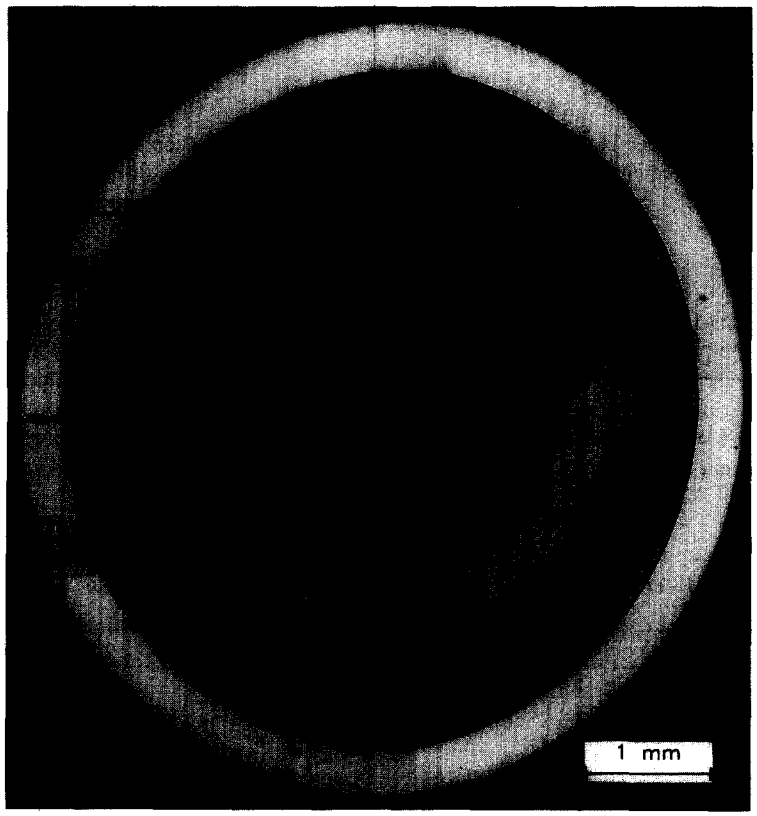


Fig. 2. Metallographic cross section with superimposed radial microprobe scans at top of fuel element DP-81 at 5% peak burnup (as polished).

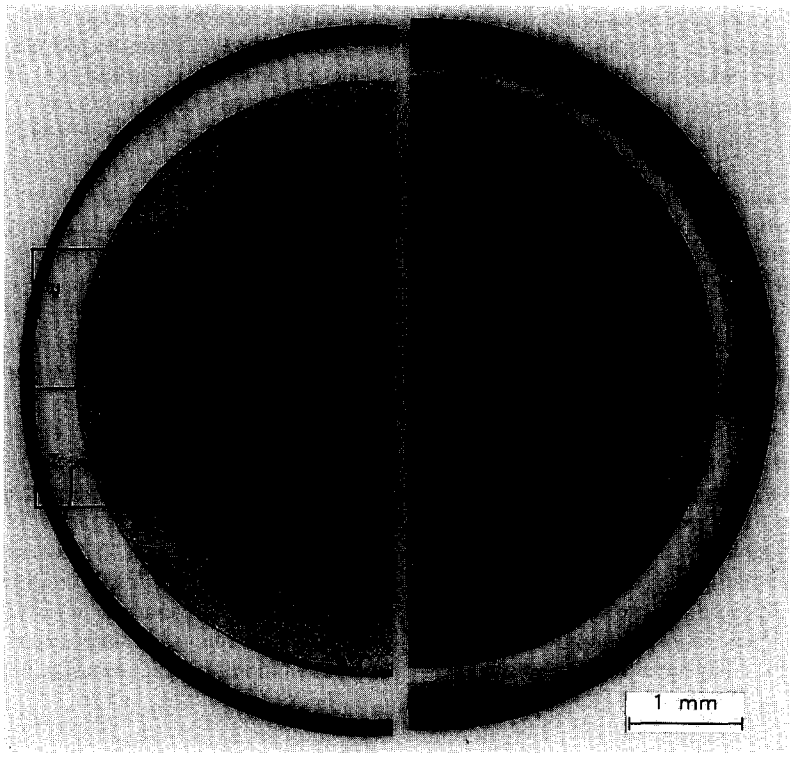


Fig. 3. Metallographic cross sections (*etched*) at 0.36 m from bottom of fuel pin, DP-11, at 10 at% peak burnup (*left*), and 0.30 m from bottom (*right*) with superimposed radial microprobe scan at 0.36 m. Increase in diameter of central γ-phase zone at 0.36 m indicates higher fuel temperature toward top of fuel pin.

ature profile in fuel pins [6]. Evaluating this possibility is an important motivation for the modeling effort described in the subsequent sections.

3. Model development

The mass flux of a species subject to potential gradients can be calculated using the phenomenological equations of non-equilibrium thermodynamics and Onsager's reciprocity theorem [7]. To simplify the model, however, it is assumed that (1) the cross-terms in the phenomenological equations are negligible, and (2) the intrinsic uranium flux in the interdiffusion region is equal to the negative of the intrinsic zirconium flux (i.e., the fluxes are observed in a fixed reference system represented by the fuel element cladding and the radial centerline of the fuel pin). This assumption is justified by the observation that once the fuel contacts the cladding early during the irradiation period (i.e., by approximately 0.5 at% burnup), the fuel and cladding remain in contact during further irradiation and no void at the fuel pin centerline develops (Fig. 2 and Fig. 3).

In the composition and temperature ranges of interest both single-and dual-phase fields exist (Fig. 5). Within a

Table 1
Composition in at.% of bands in U–Zr sample A/G 395A2 (10% burnup). Each value is the average of measurements taken at five similar locations

Description	Phase	Uranium	Zirconium
Fuel central zone	γ	52	48
Intermediate, dense zone			
High-Z major phase	β	99	~ 1
Low-Z minor phase	γ_2	60	40
Outer zone			
High-Z major phase	α	99	~ 1
Low-Z minor phase	γ_2	48	52

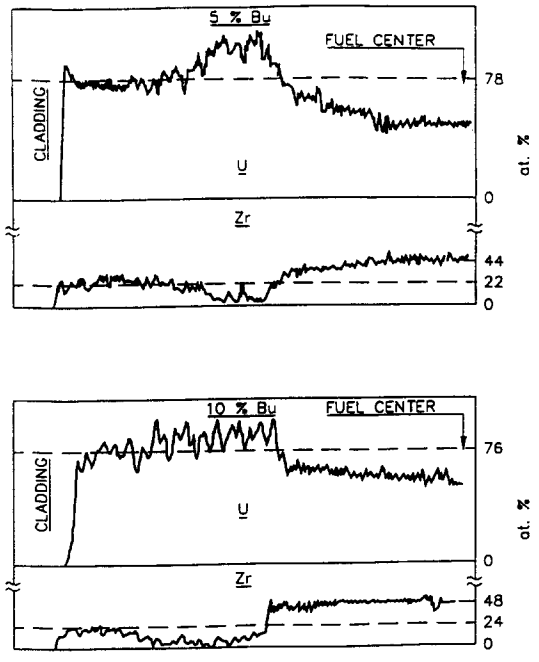


Fig. 4. Radial microprobe scans at top of fuel pins DP-81 (5 at% peak burnup) and DP-11 (10 at% peak burnup).

single-phase field, the Zr interdiffusion flux is calculated as

$$\tilde{J}_{Zr} = -\tilde{D}^* \left(\frac{\partial C}{\partial r} + \frac{\tilde{Q}^* C}{RT^2} \frac{\partial T}{\partial r} \right), \tag{1}$$

where C is the Zr concentration (mol/m^3), R is the universal gas constant (J/mol · K), and r denotes the radial dimension in the cylindrical fuel pin. $\tilde{D}$ is the U–Zr chemical or interdiffusion coefficient (m^2/s), defined by

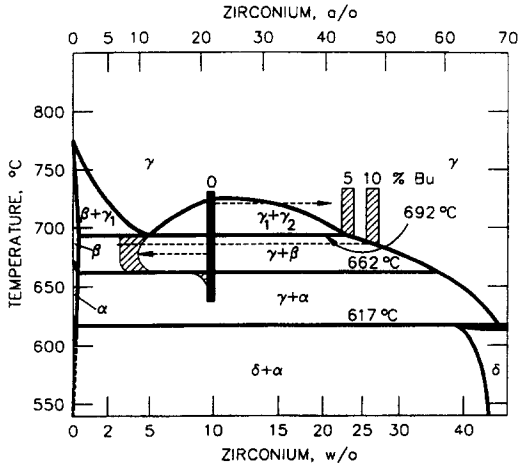


Fig. 5. Partial U–Zr phase diagram with approximate radial composition at the top of fuel pins at 0, 5, and 10 at% peak burnup.

the Darken equation as

$$\tilde{D} = X_{Zr} D_U + X_U D_{Zr}, \tag{2}$$

where X_{Zr} and X_U are the Zr and U mole fractions and D_{Zr} and D_u are the intrinsic diffusivities.

The thermo-transport parameter $\tilde{Q}^*$ (J/mol) is the apparent process heat of transport, defined as the heat associated with the redistribution of one mole of Zr or U in the fixed-reference system [12].

In a dual-phase field, the approach of Shewmon [8] can be employed. The validity of Shewmon's treatment has been verified by several experiments with dilute alloys; carbon in Fe and H in Zr, Shewmon [8]; Co in Fe–Ni, Okafar et al. [9]; and Co in Th, Axtell and Carlson [10]. Assuming conditions of local equilibrium, the concentration of Zr, C, in the γ_2-phase varies with temperature as

$$C = C_0 \exp\left[-\frac{\overline{\Delta H_s}}{RT} \right], \tag{3}$$

where $\overline{\Delta H_s}$ is the partial molal enthalpy of solution (J/mol) of Zr in U. It is apparent in the phase diagram, shown in Fig. 5, that the concentration of Zr in the γ_2-phase decreases with increasing temperature in the two phase regions. This, in turn, represents a radial concentration gradient since temperature varies with radius. Differentiation of Eq. (3) yields:

$$\frac{\partial C}{\partial r} = C \times \frac{\overline{\Delta H_s}}{RT^2} \times \frac{\partial T}{\partial r}. \tag{4}$$

Substitution of Eq. (4) into Eq. (1) gives the expression for calculating the Zr interdiffusion flux in a dual-phase field:

$$\tilde{J}_{Zr} = -\tilde{D} C \frac{V_f \left(\overline{\Delta H_s} + \tilde{Q}^* \right)}{RT^2} \frac{\partial T}{\partial r}, \tag{5}$$

where the volume fraction of the phase of interest, V_f, has also been introduced.

These flux equations were implemented within a finite-difference based computer program for solving the diffusion equation in radial geometry. The computer program discretizes the radial dimension of the fuel (Fig. 6), where R is the fuel radius. For the general node P at time t, the concentration of Zr is calculated as

$$C_P = C_P^0 + 2\Delta t \frac{\left[r_W J_W - r_E J_E \right]}{\left[r_E^2 - r_W^2 \right]} + S\Delta T, \tag{6}$$

where C_P is the Zr concentration at time t (mol/m^3), C_P^0 is the Zr concentration at the previous time-step $t - \Delta t$ (mol/m^3), Δt is the time-step size (s), r is the radial distance from the fuel center (cm), J is the Zr flux (mol/m^2 s), and S is the volumetric source of Zr as a

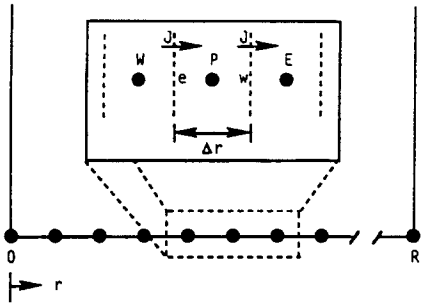


Fig. 6. Computer model discretization scheme.

stable fission product (mol/m^3 s). Explicit time-marching is employed, and the time-step size is calculated to be

$$\Delta t < \frac{(\Delta r)^2}{2D}, \tag{7}$$

for stability. The temperature profile within the fuel is computed on the same numerical grid given a fuel surface temperature and linear heat generation rate; the heat generation is radially distributed in proportion to the fissile uranium density, which evolves with time. It was observed from the computer model, however, that the temperature profile changes slowly during the Zr redistribution, so the temperature profile can be updated using time-steps much larger than those used for the diffusion calculation.

The computer program saves the average Zr concentration for each node. Polynomial equations were developed to describe the phase boundaries shown on the partial phase diagram in Fig. 5. For each time-step and each node, the computer program determines the phase field within which the node lies from the average Zr concentration and temperature of the node. If two phases are present at a node, the computer program computes the mole and volume fractions of each phase, evaluates material properties for both phases, and computes the Zr flux, $\tilde{J}_{Zr}$, using Eq. (5) for both phases. An option exists within the computer program to discontinue computing the Zr flux through a phase if its volume fraction falls below a user-defined limit.

4. Model parameters

4.1. Literature values

The parameters required by the model are material properties of the δ, α, β, and γ-phases. Specifically, values or equations for the interdiffusion coefficients, activity coefficients, and heat of transport of each phase are needed for the analysis.

Since none of the required data exist for the intermediate δ-phase, the model can either apply properties equal to those for the α-phase, or no diffusion is tracked in the δ-phase. For this reason, the model is not considered reliable for temperatures below 890 K, where the δ-phase is present. This is of no great concern as no discernable constituent redistribution appears to occur in the binary alloy below this temperature.

For the β-phase, a diffusion couple experiment was recently performed between pure uranium and a U–20 at% Zr alloy at 1023 K [11]. The average effective interdiffusion coefficient was determined to be 6.2×10^{-8} m^2/s for the β-phase. Since the activation energy for this phase is unknown in the binary system, this study will apply the activation energy for β-U self-diffusion [12].

U–Zr interdiffusion data are unavailable for the α-phase, so the diffusion coefficient was assumed to be equal to the uranium self-diffusion coefficient in this phase [13–15]. The α and β-phases are uranium phases with low solubilities for Zr (Fig. 5) and are treated as Henrian solutions [16]. Thus, the computer program treats the partial molal enthalpy of solution in the α and β-phases as negligibly small, compared to that in the high-Zr-phase. The heats of transport for these phases are considered to be model variables.

Table 2
Nominal model parameters employed in the computer program

Parameter	Phase		
	α	β	γ
Diffusion coefficients:			
D_0 (m^2/s)	2.0×10^{-3}	5.7×10^{-1}	$\log D_0 = -2.67 - 8.05X_{Zr} + 9.13X_{Zr}^2$
Q (kJ/mol)	170	180	$Q = 128 - 107X_{Zr} + 174X_{Zr}^2$
Activity coefficients:			
(kJ/mol)	negligible	negligible	$RT \ln(\gamma_{Zr}) = X_U^2(43837 - 22T - 44206X_{Zr} + 9234X_{Zr}^2)$
Heat of transport	unknown	unknown	−50 to −100 kJ/mol

$\tilde{D}$ $= D_0 \exp(-Q/RT)$ = diffusion coefficient
R = gas constant
T = temperature (K)
X_{Zr} = Zr mole fraction
Z_U = U mole fraction

The bcc γ-phase (γ-U, β-Zr) is by far the most widely studied phase. The computer program uses temperature- and Zr content-dependent interdiffusion coefficient equations extracted from the U–Zr interdiffusion data of Adda et al. [13]. Furthermore, the relationship for the Zr activity coefficient (γ) given by Leibowitz et al. [16] is used to calculate the partial molal enthalpy of solution $\overline{\Delta H_s}$:

$$\overline{\Delta H_s} = -RT^2 \frac{\partial \ln \gamma}{\partial T}. \tag{8}$$

Although the heat of transport of the phase is considered to be a model variable, an estimate of its magnitude can be made from the electron microprobe traces given in Fig. 4 and will be discussed in the next section. Table 2 summarizes the material properties used in the computer program. It should be emphasized that the diffusion coefficients used in this study are approximations based on limited data.

4.2. Estimation of heat of transport

The heat of transport for the γ-phase can be estimated, without the use of diffusion coefficients, from the microprobe trace taken after irradiation to a peak burnup of 10 at%, if it is assumed that the Zr-concentration profile has essentially reached its steady-state (or the continuing mass flux is negligibly small). This appears to be a reasonable assumption since the profile in the γ-phase at 10 at% burnup is not drastically different from the profile taken at 5 at% burnup. If the γ-phase material (central zone) has reached a steady-state (i.e., $\tilde{J}_{Zr} \approx 0$), Eq. (1) can be manipulated to solve for the heat of transport as

$$\tilde{Q}^* = R \frac{\partial \ln C_{Zr}}{\partial\left(\frac{1}{T}\right)}. \tag{9}$$

Rather than measuring the slope of the Zr concentration gradient from the Zr trace at 10 at% burnup, it was inferred from the U trace which has a more discernible slope. The temperature gradient in the inner zone, however, is subject to a considerable degree of uncertainty. The temperature at the intermediate and central zone boundary is assumed to be 965 K (from the phase diagram). The inner zone is a highly-porous region. These pores are presumably interconnected and may communicate with the gas and sodium containing plenum above the fuel pin. The model of Bauer [18] for the effective conductivity of a medium containing sodium-filled pores was employed by Yacout et al. [6] to estimate the fuel centerline temperature for this element. They obtained a centerline temperature of 976 K assuming the pores were fully saturated with sodium, and a temperature of 1003 K assuming the pores were nonconducting. This variation in the temperature gradient leads to estimates for $\tilde{Q}^*$ that range from −100 to −200 kJ/mol.

For a comparison, we may use Bober and Schumacher's [1] definition of the process heat of transport:

$$\tilde{Q}^* = \frac{D_{Zr} Q^*_{Zr} - D_U Q^*_U}{\tilde{D}}. \tag{10}$$

With the diffusion coefficients derived from Adda's experiments [13] that are applicable to the composition and temperature in the Zr-enriched γ-phase of the center of the fuel pin (Table 2), and values for $Q^*_U \approx +20$ kJ/mol from D'Amico [19] and $Q^*_{Zr} \approx -120$ kJ/mol from Campbell [20], we obtain $\tilde{Q}^*\gamma \approx -80$ kJ/mol, a value, perhaps fortuitously, close to our lower estimate of −100 kJ/mol, considering the uncertainties in all the coefficients.

5. Model calculations

The computer model described above was used to make calculations based on the conditions experienced by elements DP-81 and DP-11 from experimental subassembly X447, for which the electron microprobe examination results are shown in Fig. 4. Element DP-81 was examined at a peak element burnup of 5 at%; the microprobe trace shown in Fig. 2 and Fig. 4, however, was taken at an axial elevation 32.6 cm from the fuel bottom ($z/L \approx 0.95$), which was calculated to have a local burnup of 3.6 at%. Similarly, for element DP-11, which was examined after reaching a peak element burnup of 10 at%, the local burnup at the elevation of the examination shown in Fig. 3 (31.9 cm or $z/L \approx 0.93$) is 7.7 at%.

The temperature profile given by Yacout et al. [6] for this element DP-81 at the axial location of the microprobe examination was used as a reference for these calculations; the beginning-of-life (BOL) fuel surface temperature and linear power were assumed to be 900 K and 24000 W/m (240 W/cm), respectively. For DP-11, a fuel surface temperature of 910 K and linear power of 25000 W/m

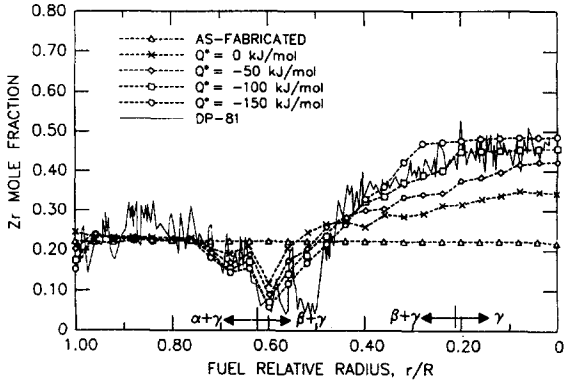


Fig. 7. Model calculations compared to microprobe results near top of fuel column for element DP-81 from experiment X447 (3.6 at% local burnup).

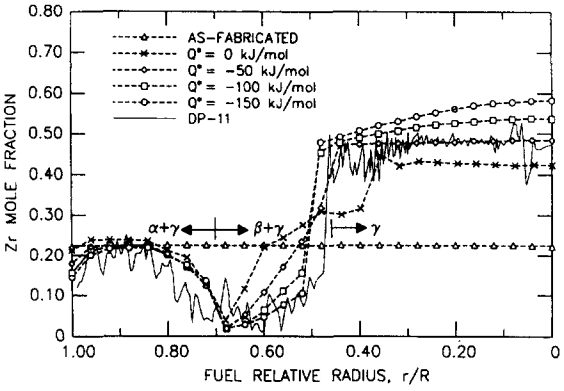


Fig. 8. Model calculations compared to microprobe results near top of fuel column for element DP-11 from experiment X447 (7.7 at% local burnup).

(250 W/cm) were used. The power was distributed radially in proportion to the radial U concentration. During the calculations, the temperature profile was allowed to evolve with the redistribution of the U and Zr constituents, buildup of porosity and subsequent ingress of bond sodium, and the reduction in element power due to fuel burnup. The fully dense U–Zr thermal conductivity was multiplied by a porosity correction factor that changed with burnup to empirically track the evolution of the fuel porosity and eventual ingress of some of the bond sodium. This porosity correction factor decreases from unity at the BOL to a minimum of 0.5 by 0.75 at% burnup, and then increases linearly to 0.7 by 1.5 at% burnup; above 1.5 at% burnup the porosity correction factor remains at the constant value of 0.7. The empirical algorithm is based upon the experimental observations of the changes in metallic fuel thermal conductivity, as reported by Betten [17]. This porosity correction factor was applied uniformly across the radial cross section of the fuel. The resulting calculated peak fuel centerline temperatures were in the range of 993 to 1008 K and occurred around 0.75 at% burnup, which are in reasonable agreement with the more detailed calculations of Yacout et al. [6]. The detailed temperature calculations of

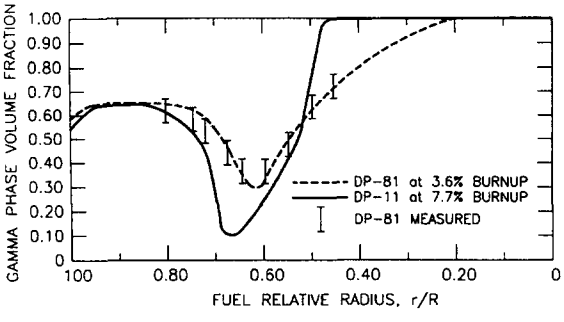


Fig. 9. Evolution of the radial distribution of the γ-phase volume fraction.

Yacout et al. utilized measured porosity distributions of these same element cross sections and a mechanistic model for determining local variations of thermal conductivity [18]. Calculations were performed with out-of-pile diffusion coefficients and thermodynamic parameters (Table 2); however, the extent of the calculated constituent redistribution turned out to be negligibly small.

It was necessary to enhance the diffusion coefficients for all phases (α, β, and γ), as calculated by the equations given in Table 2, uniformly by a factor of 10 to yield the kinetic response matching the experimental observations. The results of the two calculations are shown in Figs. 7 and 8. In these two figures, the heat of transport of the γ-phase ($\tilde{Q}^*$) is varied parametrically between 0 and −150 kJ/mol. A value of $\tilde{Q}^* \approx -50$ to -100 kJ/mol appears to produce good agreement between the model calculations and the microprobe results. Fig. 9 shows in more detail the calculation results for $\tilde{Q}^* \approx -100$ kJ/mol.

6. Discussion

Careful analysis of the results of the model calculations, displayed in Fig. 7, Fig. 8 and Fig. 9, provides interesting insights into the nature of the redistribution mechanism. Moreover, the model calculations aid in the interpretation of the microprobe results, specifically the qualitative and quantitative differences between the microprobe traces for DP-81 versus DP-11.

Fig. 7 shows the results of the model calculation for element DP-81 at 3.6 at% local burnup. Using a $\tilde{Q}^*$ value of zero clearly underpredicts the amount of radial redistribution, though there is still significant Zr migration due to the concentration gradients dictated by the phase diagram. The calculation using $\tilde{Q}^* \approx -100$ kJ/mol yields the best agreement with the DP-81 microprobe trace. The uncertainty on this heat of transport value, primarily due to uncertainties in the calculated temperatures, should be regarded as ±40 kJ/mol. The model calculation predicts that three-phase fields exist across the fuel radius at this burnup, an -s

it wasannular α + γ zone on the periphery of the fuel, and intermediate β + γ zone, and a central zone of pure γ, as indicated in Fig. 7 for the $\tilde{Q}^* \approx -100$ kJ/mol calculation. The redistribution calculation for this element, examined at low burnup, was found to be very sensitive to the way in which the fuel temperature is calculated early in the element's life. The burnup dependence of the thermal conductivity, described in the previous section, was found to yield the best calculation results when compared to the experimental data. At the terminal burnup of 3.6 at%, the apparent zone boundary at $r/R \approx 0.20$ is the β + γ ↔ γ phase boundary and does not correspond exactly to 965 K The calculation indicates the temperature of this zone boundary is ~957 K somewhat to the right of the triple

point in the phase diagram, but not a large discrepancy considering the computational and experimental uncertainties. The apparent zone boundary at $r/R \approx 0.64$ is the $\alpha + \gamma_2 \leftrightarrow \beta + \gamma_2$ phase boundary that occurs at 935 K.

Fig. 8 shows the results of the model calculation for element DP-11 at 7.7 at% local burnup; a $\tilde{Q}^*$ of -50 kJ/mol yields the best agreement with the DP-11 microprobe trace. The estimated uncertainty in this heat of transport value is ± 25 kJ/mol. Again, thermotransport is necessary to account for the amount of redistribution. Since this fuel element went to a much higher burnup, the calculation is much less sensitive than the previous calculation to the temperature variations that occur during the first two percent burnup. For these conditions the model calculation also predicts that three phase fields exist across the fuel radius, an annular $\alpha + \gamma$ zone on the periphery of the fuel, an annular intermediate $\beta + \gamma$ zone, and a central zone that is pure γ. Shown on the figure are the calculated locations of the phase transitions for $\tilde{Q}^* \approx -100$ kJ/mol. The calculation indicates that the apparent zone boundary at $r/R \approx 0.46$ is again the $\beta + \gamma \leftrightarrow \gamma$ phase boundary (calculated to be ~ 940 K. The apparent zone boundary at $r/R \approx 0.72$ is, as before, the $\alpha + \gamma_2 \leftrightarrow \beta + \gamma_2$ phase boundary that occurs at 935 K.

Fig. 9 shows the calculated γ-phase volume fractions across the fuel radii at the local burnup at the top of both DP-81 and DP-11. The calculation for DP-81 compares well with the measurements made on the metallographic section shown in Fig. 2.

As mentioned before, it proved impossible to model the observed U–Zr redistribution with independently measured (ex-reactor) diffusion data for pure U–Zr alloys. However, a tenfold increase in all interdiffusion coefficients results in good agreement between model results and postirradiation data. Such a large increase cannot be justified by scatter in the experimental diffusion data. Although there are considerable differences in the measured diffusion coefficients for compositions at the extremes of the phase diagram between the experiments of Adda [21] and Müller [22] and those of Fedorov [23], in the Zr concentration range of 20 to 60 at.% all diffusion data, including those of Petri [11], agree within a factor of two. It has been shown by Peterson [12] that impurities affect the self-diffusivity of uranium. For example, fast diffusing elements like Co and Fe increase the U diffusion coefficient, whereas Nb –a slow diffuser –has the opposite effect. Considering the large variety of fission product elements present, an evaluation of their specific role appears quite intractable without further experimentation.

In addition to fission products, Pu is produced by neutron capture in ^{238}U, at a rate of 0.05 at% per % burnup. Dupuy [24] and Petri [25] have shown that Pu greatly enhances the inter-diffusivity in U–Pu alloys, even at small concentrations. Whether this effect can be extended to U–Zr alloys is not certain in the absence of experimental data, though experiments in ternary U–Pu–Zr alloys with 20 at% Pu have γ-phase diffusivities an order of magnitude greater than those in binary U–Zr [11].

It is interesting to note that in a related study of swelling behavior of the present U–Zr alloys, Rest [26] also needed to increase diffusion coefficients in order to model the observed fission gas bubble morphology. The standard irradiation enhanced diffusion mechanisms do not yield a significant enhancement because at the high temperature prevailing in the diffusion zones thermally-activated vacancies overwhelm irradiation-induced point defects. Rest was able to calculate a tenfold increase of the diffusivity by invoking a mechanism proposed for irradiation-induced solute segregation. The diffusing entity in this mechanism is an irradiation–induced Zr interstitial pair. In order for this to work, a rather large number of interstitials have to survive recombination with vacancies. This can be accomplished by providing very effective vacancy sinks. These sinks may indeed exist in the form of a high number density of small fission gas bubbles.

7. Conclusions

Analysis of electron microprobe examinations of two U–Zr fuel elements having peak burnups of 5 and 10 at%, coupled with the development and validation of a mechanistic model, leads the authors to the following conclusions regarding constituent redistribution in U–Zr fast reactor fuel:

(1) There exists a driving force for Zr to move up the temperature gradient owing to a negative $\overline{\Delta H_s}$ for the γ-phase where it exists in the $\alpha + \gamma$, $\beta + \gamma$, and $\gamma_1 + \gamma_2$-phase fields; Zr depletion occurs primarily in $\beta + \gamma$, rather than in $\alpha + \gamma$, because of a low-Zr mobility in the α-phase.
(2) There is strong evidence, from experimental as well as computational results, for a heat of transport of Zr in the γ-phase (a γ-U, β-Zr solid solution) in the range of -50 to -100 kJ/mol; chemical interdiffusion alone could not account for the extent of redistribution.
(3) As no discernable driving force for Zr transport down the temperature gradient appears to exist, a Zr depleted fuel center will not develop.
(4) The transport model presented here can only match experimental observations when ex-reactor diffusion coefficients are assumed to be enhanced by irradiation effects.

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References

[1] M. Bober and G. Schumacher, in: Advances in Nuclear Science and Technology, vol. 7, ed. E.J. Henley and J. Lewis (Academic Press, New York 1973) p. 121.
[2] W.F. Murphy, W.N. Beck, F.L. Brown, B. Koprowski and L.A. Neimark, Postirradiation Examination of U–Pu–Zr Fuel Elements Irradiated in EBR-II to 4.5 Atomic Percent Burnup, ANL-7602, Argonne National Laboratory (1969).
[3] R.G. Pahl, D.L. Porter, C.E. Lahm and G.L. Hofman, Metall. Trans. 21A (1990) 1063.
[4] M. Ishida, T. Ogata and M. Kinoshita, J. Nucl. Mater. 104 (1993) 21.
[5] T. Ogawa and T. Iwai, J. Less-Common Metals 175 (1991) 59.
[6] A.M. Yacout, G.L. Hofman, Y. Orechwa and S. Salvatores, Trans. Am. Nucl. Soc. 68 (1993) 98.
[7] S.R. de Groot, Thermodynamics of Irreversible Processes (North-Holland, Amsterdam, 1952).
[8] P.G. Shewmon and D. Jaffe, Acta Metall. 12 (1964) 515.
[9] I.C.I. Okafar, O.N. Carlson and D.M. Martin, Metall. Trans. 13A (1982) 1713.
[10] S.C. Axtell and O.N. Carlson, Metall. Trans. 21A (1990) 2141.
[11] M.C. Petri, Dissertation, Purdue University (April 1995), to be published.
[12] N.L. Peterson and S.I. Rothman, Diffusion in Gamma Uranium, ANL-6568, Argonne National Laboratory (1965).
[13] Y. Adda and A. Kirianenko, J. Nucl. Mater. 6 (162) 135.
[14] S.J. Rothman, L.T. Lloyd and A.L. Harkness, Trans. Am. Inst. Mech. Eng. 218 (1960) 604.
[15] G. Martin and B. Perraillon, Grain Boundary, Structure and Kinetics (American Society of Metals, Metals Park, OH, 1980) p. 239.
[16] L. Leibowitz, R.A. Blomquist and A.D. Pelton, J. Nucl. Mater. 167 (1989) 76.
[17] P.R. Betten, Trans. Am. Nucl. Soc. 50 (1985) 239.
[18] T.H. Bauer, Int. J. Heat Mass Trans. 36 (1993) 4181.
[19] J.F. D'Amico and H.B. Huntington, J. Phys. Chem. Solids 30 (1969) 1697.
[20] D.R. Campbell and H.B. Huntington, Phys. Rev. 179 (1969) 601.
[21] Y. Adda, J. Philibert and H. Faraggi, Revue De Metallurgie 54 (1957) 597.
[22] N.Z. Müller, Z. Metallk. 50 (1959) 562.
[23] G. Fedorov, E.A. Smirnov and F.I. Gomov, Metallurgia and Metalovdenie 7 (1968) 116.
[24] M. Dupuy and D. Calais, Comp. Rend. Acad. Paris 260 (1965) 1412.
[25] M.C. Petri, A.G. Hins, J.E. Sanecki and M.A. Dayananda, J. Nucl. Mater. 211 (1994) 1.
[26] J. Rest, J. Nucl. Mater. 207 (1993) 192.